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Microwave-assisted one-pot synthesis of carbon dots for highly sensitive and selective detection of selenite

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ABSTRACT

Herein, a label-free, sensitive, and selective fluorescence method for the detection of trace selenite (Se(IV)) in water sample by carbon dots (CDs) was reported. The fluorescence CDs probe was simply and rapidly synthesized within 4 min by microwave-assisted pyrolysis of citric acid and 3,3′-diaminobenzidine (DAB), with a relatively uniform size of approximately 2.6 nm. The fluorescence emission of as-prepared CDs was 450 nm upon the excitation of 400 nm with a quantum yield of about 8% against quinine sulfate as reference standard, which can be quenched by the addition of selenite. Benefit from the strong surface chelating reaction between DAB and Se (IV), the proposed sensor owned the advantage of high sensitivity (limit of detection = 0.5 ng/mL), wide linear range (1–100 ng/mL range, relative standard deviation: 3.8%), and excellent selectivity in the presence of various ions, even other forms of selenium, including selenate (Se(VI)), Selenomethionine (SeMet), and L-Selenocystine (SeCys) etc. Preliminary study on fluorescence quenching mechanism studied. The sensor was successfully exemplified for real water samples and exhibits comparative performance.

1. Introduction

Selenium (Se) is a natural element widely found in soil, water and plants[1]. Meanwhile, Se is an essential micronutrient, playing an important role in human health serving as antioxidant, immunoregulatory, antagonistic roles and as well as other biological functions.[2] Low selenium status is linked to Keshan disease (KD) and white muscle disease[3]. Therefore, proper supplementation of selenium in selenium-deficient areas is beneficial to health. However, only a narrow range of selenium concentrations is beneficial,[4] the recommended intake for adults is 55 μg/day and the tolerable upper is 400 μg/day[5]. Excessive uptake of Se can be toxic, leading to oxidative stress and other toxicity effect.[6] The recommend level of Se by the World Health Organization (WHO) in drinking water is less than 10 ng/mL. Furthermore, the toxicity of Se depends on its chemical form. The common chemical form of Se in natural water is element selenium (Se^0), selenide (Se^{2-}), selenite (Se(IV)), and selenate (Se (VI))[7]. Among them, Se(IV) is much more harmful than the others[8]. Therefore, the accurate and sensitive determination of Se (IV) in water resources is important to reduce threats to human health.

Over the years, lots of techniques for the determination of Se(IV) have been reported, including fluorescence spectrometry (FL)[9,10], UV–visible spectrophotometry (UV–Vis)[11,12], gas chromatography (GC)[13,14], high-performance liquid chromatography (HPLC),[15] atomic absorption spectroscopy (AAS),[16] atomic fluorescence spectroscopy (AFS)[17,18], and Inductively coupled plasma mass spectrometry (ICP-MS)[19]. Among them, FL is gaining interest due to its advantage of simple operation, low cost, visualization and the potential of instrument miniaturization[20–22]. Recently, there are many FL probes based on nanoparticles have been developed[23]. For instance, silica nanoparticle coded 3,3′-diaminobenzidine(DAB) was reported by Liang et al.[24] to detect Se(IV) on the fluorescent turn-on mode. Owing to the strong chelating reaction between DAB and Se(IV), selective FL sensing was implemented and a detection limit (LOD) of 0.19 μg/mL was

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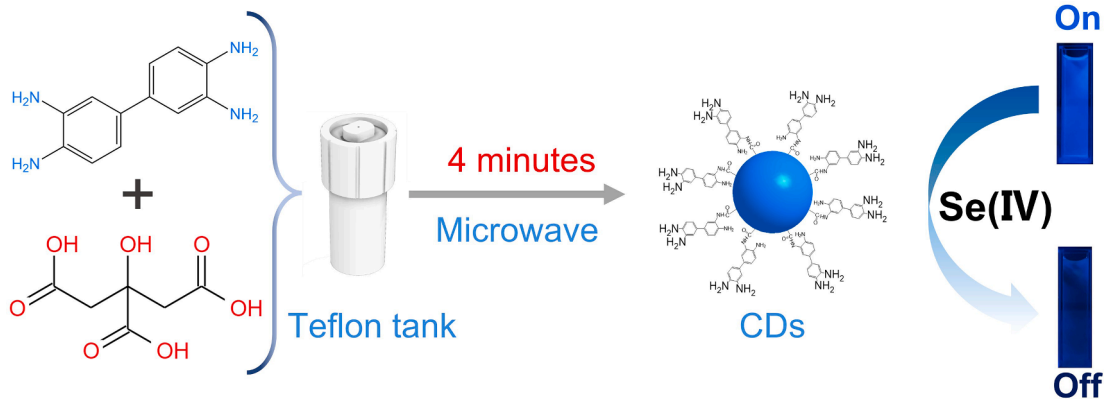

Scheme 1. Microwave assisted synthesis of CDs and its FL sensing towards Se(IV).

achieved. To further improve the sensitivity and inherent stability of FL method, Chen et al.[25] designed a ratiometric fluorescent nanosensor by linking DAB onto the surface of silica nanospheres embedded with CdTe QDs. The ratio of two emission peaks linearly increased with the increasing of Se(IV) concertation and the detection limit reached as low as 0.53 ng/mL. Huang et al.[26] prepared a water-soluble CdS quantum dots capped by mercaptoacetic acid. The selective detection for Se(IV) was achieved in the presence of glutathione (GSH) with the LOD of 0.03 mmol/L. To avoid the use of toxic inorganic QDs, Devi et al.[27] prepared carbon dots(CDs) as luminophore coating with nitrogen-rich supramolecular ligand as identifier. The prepared CDs showed selective quenching in PL intensity by Se(IV) and exhibited a theoretical LOD of 0.1 ng/L. Meanwhile, a major dairy waste, whey was used to synthesis CDs by Devi's group[28], and the obtained CDs also showed high sensitivity to Se(IV) over other relevant ions. Despite the excellent performance of these reported nanoprobes, their long synthesis time and the complex post-modification limit the widespread adoption of these methods.

To avoid time-consuming and complex post-modification, easy and fast one-pot method was selected for the preparation of CDs[29,30]. As showed in Scheme 1, citric acid (CA) and DAB were dissolved in water and then fluorescent CDs were obtained via microwave heating within 4 min, showing bright fluorescence emission at 450 nm under 400 nm excitation. The prepared CDs are dispersed spherical carbon

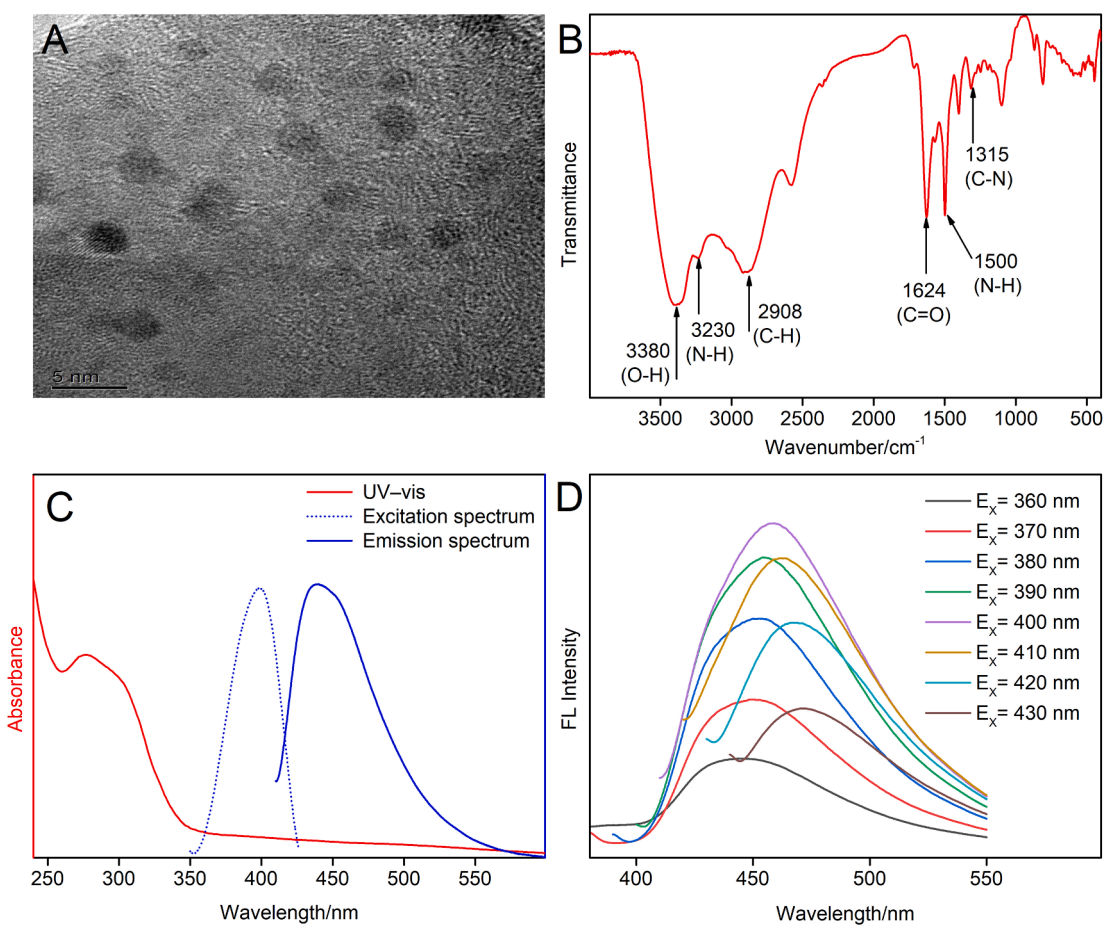

Fig. 1. TEM image (A), FTIR spectrum (B), UV–vis absorption spectrum, excitation spectrum, emission spectrum (C) and fluorescence emission spectra under different excitation wavelengths (D) of the prepared CDs.

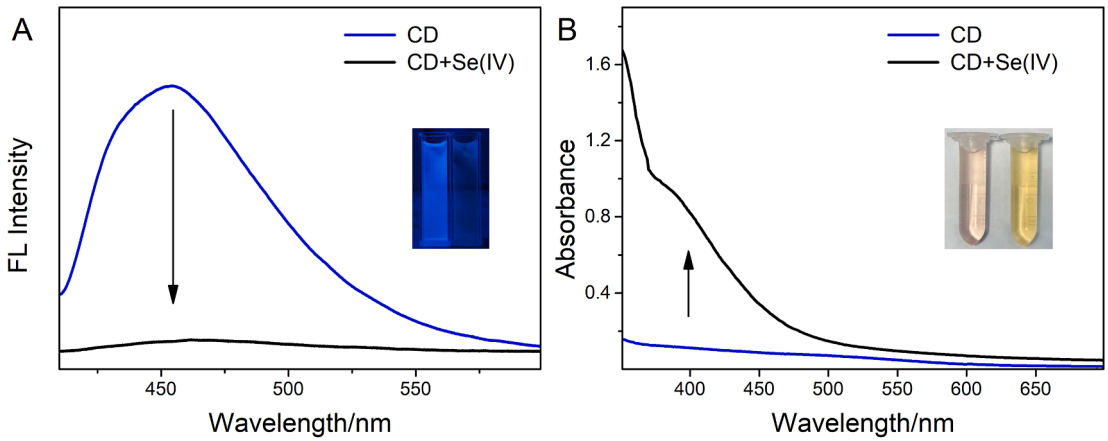

Fig. 2. The (A) FL spectra and (B) UV–vis spectra of CDs with the addition of Se(IV). The concentration of Se(IV): 10 μg/mL.

nanoparticles with a diameter distribution ranging from 2 to 3 nm and an average of ~ 2.6 nm. As the result of the selective strong chelate reaction between DAB and Se(IV), the fluorescence of CDs can be quenched with the addition of Se(IV) to achieve selectively and sensitively determination in aqueous system. Moreover, the developed method was further explored to real sample analysis, with good results obtained for the determination of both Se(IV) contained in natural water samples.

2. Experimental

2.1. Chemicals and Instruments

All reagents used throughout this work were at least analytical grade or better. Ultrapure water (18.2 MΩ·cm) was produced with a purification water system (PCWJ-10, Pure Technology Co. Ltd, Chengdu, China). Potassium Selenate (Se(VI)) was purchased from Alfa Aesar Co. Inc. (Shanghai, China), while Sodium Selenite (Se(IV)) was obtained from Sigma-Aldrich (Darmstadt, Germany). Other reagents were obtained from Aladdin Reagents Co. Ltd. unless specified. The stock solutions (1000 μg/mL) of Se (IV) were prepared by dissolving an appropriate amount of sodium selenite in ultrapure water respectively. All the standards and stock solutions were stored at 4 °C in a refrigerator until use.

Microwave oven (P70D20TL-D4, Galanz Enterprises Co., Ltd., Guangdong, China) was used to synthesis CDs. Transmission electron microscopy (TEM) image was collected on a FEI Tecnai G2 F20 electron microscope operated at 200 kV. WQF-520A FTIR spectrometer (Beijing Beifen Ruili Analytical Instrument Co., Beijing, China) was used to collect the infrared spectrum of CDs. The fluorescence measurements were performed with a 970 CRT spectrophotometer (Shanghai Precision Scientific Instruments Co., Ltd., Beijing, China) with a quartz cell (1 cm × 1 cm), with the excitation at 400 nm. Fluorescence lifetime measurements were performed on a FLS1000 Edinburgh Analytical Instrument apparatus. Absorption spectra were recorded using a UV-1801 UV/Vis spectrophotometer (Beijing Beifen Ruili Analytical Instrument Co., Beijing, China). X-ray diffraction spectrometer (X'Pert Pro MPD, Philips, Netherlands) with Cu Kα radiation was employed to analyze the crystal structure of CDs. The X-ray photoelectron spectrum of CDs was collected by Thermo Scientific K-Alpha instrument with Al Kα X-ray source.

2.2. Preparation of CDs

For the synthesis of CDs, 0.5 mL citric acid (0.1 mol/L) and 5 mL DAB (50 mmol/L) were added in 4.5 mL H_2O. After 5 min sonication, the mixture was transferred to a sealed Teflon tank, then heated by microwave oven with relatively low power (80 W) for 4 min. After cooling to room temperature, the obtained brownish product was transferred to dialysis in a 1 kDa dialysis bag. And finally, the purified CDs were preserved at 4 °C until use.

Safety Considerations: For the safety of the operator, microwave synthesis should be carried out in proper ventilation and personal protective equipment should be employed.

2.3. Fluorescence sensing of selenite and interference studies

To demonstrate sensing capability of CDs towards Se(IV), 200 μL CDs (2.0 mg/L), 100 μL citrate buffer solution (100 mM, pH 3.0) were mixed. Then 1000 μL of serial concentrations of Se(IV) or water sample was added, and the mixture was diluted to 2.0 mL by H_2O. After equilibration for 1 h, the fluoresce spectra were recorded under the excitation of 400 nm, respectively. For interference investigation, solution containing various cations or anions were added into the mixture, followed by the addition of aqueous Se(IV) and subsequent FL measurement.

3. Results and discussion

3.1. Characterization of synthesized CDs

The size and shape of CDs were examined by TEM (Fig. 1A), it was confirmed that the as-synthesized CDs were monodispersed, sphere-shaped and the typical particle size was about 2.6 nm, the particle size distribution was shown in Figure S1. Only a broad diffraction peak at about 24° appeared in XRD spectrum (Figure S2) which is due to the amorphous nature of disordered bonded carbon atoms[28]. To get further information on the structure and components of the CDs, FTIR spectrum was performed as shown in Fig. 1B. The characteristic peak from the stretching vibration of C–H bond appeared at approximately 2908 cm^{-1}. The observed broad peak at 3380 cm^{-1} was contributed to O–H vibrational stretching indicating the residual hydroxyl groups on its surface. And the appearance of a sharp peak at 1624 cm^{-1} was assigned of the presence of C = O vibrational stretching revealed the existence of carboxylate. More importantly, the characteristic absorption peaks N–H stretching vibration (3230 cm^{-1}), N–H bending vibrations (1500 cm^{-1}), and C-N bending vibration (1315 cm^{-1}) are owing to the residual amino group from DAB. The XPS spectrum of CDs showed in Figure S3 displayed that the existence of C, O, N on the surface (Table S1). To further analysis the C 1 s peak (Insert of Figure S3), the graphitic or aliphatic (C = C and C–C), oxygenated (C = O), and nitrous carbon atoms (C-N) were discovered. These results indicated that the appearance of many functional groups (–COOH, –OH and N-containing groups) on the surface of the CDs[31], which gave the CDs excellent solubility in water without any further chemical modification.

The UV spectrum of the prepared CDs was shown in Figure 1C, a

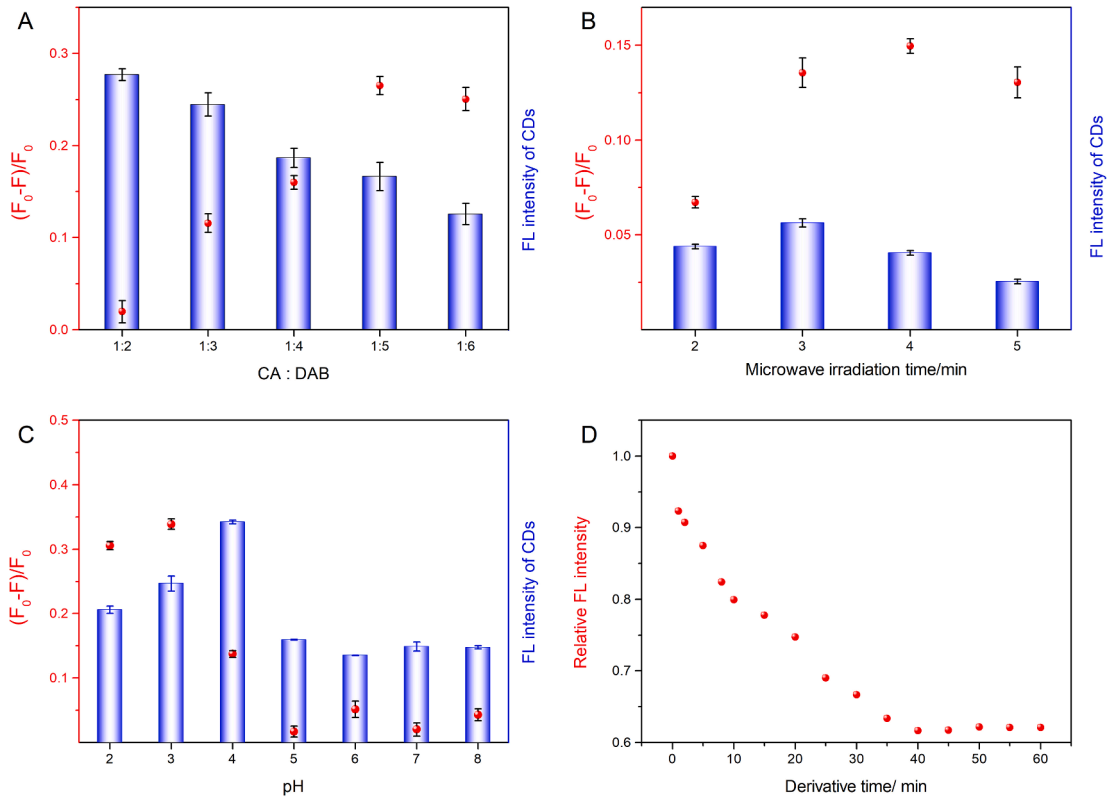

Fig. 3. FL intensity (bar) and $(F_0\text{-}F)/F_0$ (dot) of CDs at varied ratio of raw materials (A), microwave irradiation time (B), pH (C), and derivation time (D), containing Se(IV) with same concentration: 0.5 μg/mL. F_0 represents the FL intensities of CDs at 450 nm and F represents the FL intensities of CDs solution contained Se(IV).

shoulder at 240 nm mainly belong to the π-π* energy transition of the conjugated system in CDs and a peak at about 300 nm can be attributed to the n-π* transition of C = O or C = N bond[32,33]. When excited at 400 nm, the CDs showed photoluminescence in the range of 400–600 nm, with the maximum at around 450 nm (Fig. 1C), the quantum yield of CDs was about 8% against quinine sulfate as reference standard. The photograph of CDs dispersion under UV light exhibited a blue color (inset of Fig. 2A). As with most CDs, the emission spectra were sensitive to the excitation wavelength and shift of emission peaks at longer excitation wavelengths was observed in Fig. 1D, which showed its excitation dependent emission behavior and agreed with previous reports[34,35]. The influence of ionic strength on the fluorescence intensity was shown in Figure S4A. Upon addition of 1.0 M of NaCl, the FL intensity of the CDs still maintains at about 92% demonstrating the proposed CDs possess good salt interference. As shown in Figure S4B, the FL intensity maintains at about 91% after 20 min of continuous UV radiation at 400 nm, which indicates a good photostability.

3.2. Fluorescence sensing of Se(IV) by CDs

From the above structural and spectroscopic characterization, it is clear that the synthesized CDs owned the features of homogeneous in size, good water solubility, and residues with o-diamino group on the surface, which give CDs the potential of sensing Se(IV) selectively. To preliminarily validate, the FL intensity of CDs dropped with the addition of 10 μg/mL Se(IV) (Fig. 2A), the original bright blue color of CDs disappeared under the UV lamp (insert of Fig. 2A), which implies the surface moieties of CDs are able to detect Se(IV). To the preliminary study of the mechanism of the fluorescent quench, the UV–Vis absorption spectrum was investigated as shown in Fig. 2B. After the addition of Se(IV), a new absorption peak at 400 nm appeared and the color of the mixture changed into yellow, which could be owing to the formation of the ring-shaped conjugated complex between residual DAB group on the CDs surface and Se(IV)[25]. Meanwhile, the fluorescence decay curves of the prepared CDs and the react mixture of CDs and Se(IV) were shown in Figure S5. While the fluorescence lifetime did not change significantly, indicating the sensing of CDs towards Se(IV) is the inner filter effect (IFE) process[36].

3.3. The optimization of experimental conditions

To achieve the best fluorescence sensing performance, the parameters were carefully investigated, including ratio of raw materials, microwave irradiation time, pH, and derivation time. As shown in Fig. 3A, the fluorescence intensity of the obtained CDs gradually decreased (Blue bars) with the increase of DAB content, but the fluorescence sensing efficiency ($(F_0\text{-}F)/F_0$, red dots) did gradually increase. This probably because the amount of DAB groups remaining on the surface of carbon dots increased, bringing more reaction sites with Se(IV). The efficiency reached a plateau when CA:DAB was 1:5, so 1:5 was chosen as the raw material ratio for the synthesis of CDs. The biggest advantage of the microwave method is only a very short synthesis time is required. As the results shown in Fig. 2B, it only took 3 min to reach the maximum FL intensity, however, in order to pursue the maximum sensing efficiency, we finally selected 4 min as the synthesis time. The fluorescence of CDs is pH-sensitive[37], so pH of the sensing system should be studied. As shown in Fig. 3C, the FL intensity of synthesized carbon dots is higher under acidic conditions than neutral or basic, reaching a maximum at pH = 4. This is very timely for Se(IV) sensing because it is known that the reaction between Se(IV) and DAB is required under acidic conditions [38]. Finally, we found that the best efficiency of fluorescence sensing was pH = 3. Besides, the reaction time of Se(IV) and CDs is vital. When the Se(IV) was injected into the CDs solution, the fluorescence of the CDs decreased rapidly and reached equilibrium after 40 min. For insurance purpose, 1 h reaction time was chosen finally.

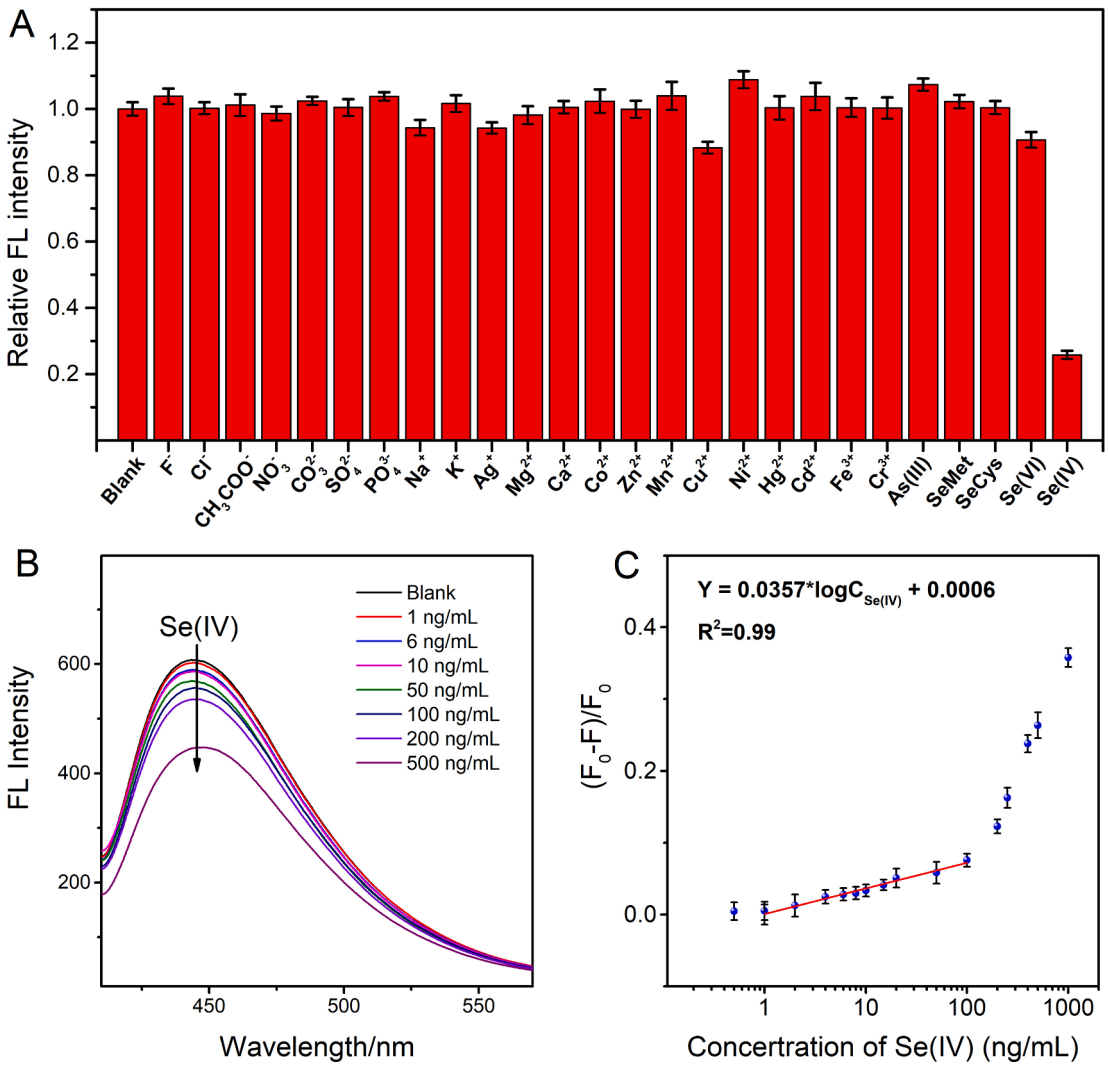

Fig. 4. A: Relative fluorescence intensity of CDs at 450 nm in the presence of different ions (excited at 400 nm). B: FL spectra of the CDs with the addition of Se(IV) (0, 1, 6, 10, 50, 100, 200 ng/mL). C: $(F_0\text{-}F)/F_0$ changing with respect to Se(IV) in 0.5 to 1000 ng/mL concentration.

3.4. Analytical performance and sample analysis

To investigate the interference of the other ions, CDs solution were titrated with various large amount of ions including F^-, Cl^-, CH_3COO^-, NO_3^-, SO_4^{2-}, PO_4^{3-}, Na^+, K^+, Ag^+, Mg^{2+}, Ca^{2+}, Co^{2+}, Zn^{2+}, Mn^{2+}, Cu^{2+}, Ni^{2+}, Hg^{2+}, Cd^{2+}, Fe^{3+}, Cr^{3+}, As(III), Se(VI), SeMet, and SeCys. The concentration of Se(IV) was 5 μg/mL, while the others were 50 μg/mL. The relative fluorescence intensity of CDs in the presence of various cations mentioned above including Se (IV) was shown in Fig. 4A. Only Se (IV) can turn off the fluorescent emission of CDs obviously, while other have not much change. This demonstrates the excellent selectivity of the developed method towards Se(IV). The observed selective performance of the probe could be attributed to the formation of the ring-shaped conjugated complex between residual DAB group on the CDs surface. Moreover, the FL response of CDs to Se(IV) in different synthetic batch was shown in Figure S6, indicating the proposed method has good reproducibility. Under the optimized conditions, by increasing the concentration of Se(IV) (0.5–1000 ng/mL), the quenching efficiency of the system gradually increased (Fig. 4B&C). And the calibration equation between 0.5 and 100 ng/mL, with the favorable correlation coefficient ($R^2 = 0.99$), was $Y = 0.0357 \times \log C_{Se(IV)} + 0.0006$, in which Y mean the quenching efficiency ($(F_0\text{-}F)/F_0$) and $C_{Se(IV)}$ was the concentration of Se(IV) in ng/mL. The theoretical limit of detection (LOD, 3σ) was 0.5 ng/mL at 3.8% relative standard deviation. Furthermore, based on the thorough comparison with the previously reported FL probes for Se(IV) (Table 1), it could be found that this developed probe herein has two advantages. Firstly, the CDs could be easily obtained without any additional modification or functionalization, which could greatly shorten the preparation time and avoid any complex synthetic

Table 1
Comparison of nano fluorescence probes for Se(IV).

Probes	Synthesis time	Modification time	Linear range (ng/mL)	LOD (ng/mL)	Ref.
DAB coated silica NPs	48 h	3.5 h	1000–10000	190	[24]
DAB coated $CdTe@SiO_2$	48 h	3 h	0–198	0.53	[25]
CdTe QD	1 h	—	5–100	0.1	[20]
CdSe QD	—	—	5–65	0.08	[39]
CdS QD	12 h	—	4–885	2.37	[26]
Nitrogen-rich CDs	15 min	24 h	0.1–80	0.1	[27]
CDs	12 h	—	10–100	0.1	[28]
Nitrogen doped CDs	12 h	—	78–21400	53	[40]
CDs	4 min	—	1–100	0.5	this work

Table 2
Analysis of water samples.

Sample	Certified value		Observed value	Recovery
BWB2261-2016	79 ± 4 ng/mL		80 ± 3 ng/mL	—
BWB247-2016	20 ± 2 μg/mL		18 ± 1 μg/mL	—
	Detected	Added		
Local lake water	ND	20.0 ng/mL	18.6 ± 0.7 ng/mL	93%
	ND	40.0 ng/mL	41.3 ± 0.5 ng/mL	103%
Local drink water	ND	40.0 ng/mL	38.4 ± 0.6 ng/mL	96%

Mean ± SD, n = 3. ND: not detected.

procedures only mastered by those with expertise. Secondly, the sensitivity was superior or at least very rivalrous to the previously reported.

To test the practical applicability of the developed method, the proposed method was employed for the analysis of Se(IV) in standard sample of selenium in seawater (BWB247-2016). As shown in Table 2, 18 μg/mL of Se(IV) was found, which was in good agreement with the certified value (*t*-test, 95% of confidence level). In addition, the standard water quality selenium sample (BWB2261-2016) was also tested with the proposed method, and the satisfied results were achieved. The developed method was also validated with a standard addition method with local lake and drink water, and the recovery of spiked Se(IV) is 93–103%. The acceptable recoveries indicated that our method has the potential in the analysis of Se(IV) in natural water samples.

4. Conclusion

In summary, we found a common and easily attainable CDs could be used as a label-free FL probe for the turn-off sensing of Se(IV) in a facile and sensitive manner. Compared with other probes, it shows the virtue of simplicity and rapid, with the sensitivity at least comparable or even better. Furthermore, the validation experiments indicated that the proposed method could be satisfactorily applied to the analysis of Se(IV) in water samples. Therefore, this developed method could show promising perspectives in real analysis for the future with high practical utility.

CRediT authorship contribution statement

Guiping Liao: Methodology, Writing – original draft, Writing – review & editing. **Jin Luo:** Writing – original draft, Writing – review & editing. **Tengfei Cui:** Data curation, Investigation, Validation. **Jie Zou:** Investigation, Methodology, Validation. **Mian Xu:** Investigation, Validation. **Yuhan Ma:** Investigation. **Lingsheng Shi:** Investigation. **Jia Jia:** Methodology, Writing – review & editing, Resources. **Chen Ma:** Investigation, Validation. **Hui Li:** Investigation. **Fujian Xu:** Conceptualization, Writing – review & editing, Supervision, Resources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.microc.2022.107440.

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